Three phase current — Solution

X23 (2023) — English translation

A1 **0.60 pts**

The line voltage between terminals a and b can be represented as $\mathcal{E}_{ab} = \mathcal{E}_l' \cos(\omega t - \Delta\varphi)$, $\mathcal{E}_l' > 0$. Find $\mathcal{E}_l'$ and $\Delta\varphi$. Express your answer using $\mathcal{E}_0$.

$$\mathcal{E}_{ab} = \mathcal{E}_a - \mathcal{E}_b = \mathcal{E}_0 \left(\cos(\omega t) - \cos\left(\omega t - \frac{2\pi}{3}\right)\right).$$

Apply formula difference cosine, obtain:

$$\mathcal{E}_{ab} = 2\mathcal{E}_0 \left(\sin\left(\omega t - \frac{\pi}{3}\right)\sin\left(-\frac{\pi}{3}\right)\right) = -\sqrt{3}\mathcal{E}_0 \sin\left(\omega t - \frac{\pi}{3}\right).$$

Replacing the sine with the required cosine:

$$\mathcal{E}_{ab} = \sqrt{3}\mathcal{E}_0 \cos\left(\omega t + \frac{\pi}{6}\right).$$

Note. in the condition the polarity is not specified $\mathcal{E}_{ab}$, therefore possible variant $\mathcal{E}_{ab} = \mathcal{E}_b - \mathcal{E}_a$. in this case sign cosine in finally expression swap, that correspond shear phase cosine on π:

$$\mathcal{E}_{ab} = \sqrt{3}\mathcal{E}_0 \cos\left(\omega t - \frac{5\pi}{6}\right).$$

Answer:

$$\mathcal{E}_l' = \sqrt{3}\mathcal{E}_0$$

$$\Delta\varphi = -\frac{\pi}{6} \quad \text{or} \quad \Delta\varphi = \frac{5\pi}{6}$$

A2 **1.00 pts**

Let each of the load elements (Z_a, Z_b, Z_c) be a resistor $R = 9$ $k\Omega$ and a capacitor $C = 2 \cdot 10^{-7}$ F connected in series. Find the current amplitudes through the line wire I_a and the neutral wire I_0 (formula and numerical value).

Let's use the method of complex amplitudes. We will denote complex amplitudes by a line over the corresponding quantities. Let's write Kirchhoff's 2nd law for a circuit with current I_a:

$$\mathcal{E}_a = \overline{I_a}(R + r) - \overline{I_a}\frac{i}{\omega C} + \overline{I_0}r;$$

Similarly for I_b, I_c:

$$\mathcal{E}_b = \overline{I_a}(R + r) - \overline{I_b}\frac{i}{\omega C} + \overline{I_0}r; \mathcal{E}_c = \overline{I_a}(R + r) - \overline{I_c}\frac{i}{\omega C} + \overline{I_0}r \tag{1}$$

Also write 1st law Kirchhoff's:

$$\overline{I_0} = \overline{I_a} + \overline{I_b} + \overline{I_c} \tag{2}$$

Express $\overline{I_a}$, $\overline{I_b}$, $\overline{I_c}$ from (1), (2), (3) correspondingly and substitute in (4):

$$\overline{I_0} = \frac{\mathcal{E}_a - \overline{I_0}r}{R + r - \frac{i}{\omega C}} + \frac{\mathcal{E}_b - \overline{I_0}r}{R + r - \frac{i}{\omega C}} + \frac{\mathcal{E}_c - \overline{I_0}r}{R + r - \frac{i}{\omega C}}.$$

Since $\mathcal{E}_a + \mathcal{E}_b + \mathcal{E}_c = 0$, we obtain

$$\overline{I_0} = 0.$$

Taking into account this have answer for $\overline{I_a}$:

$$\overline{I_a} = \frac{\mathcal{E}_0}{(R+r) - \frac{i}{\omega C}},$$

$$|\overline{I_a}| = I_a = \frac{\mathcal{E}_0}{\sqrt{(R+r)^2 + \frac{1}{(\omega C)^2}}}.$$

Answer:

$$I_a = \frac{\mathcal{E}_0}{\sqrt{(R+r)^2 + \frac{1}{(\omega C)^2}}} \approx 14{,}6\text{mA}; I_0 = 0.$$

A3 0.40 pts

Find the efficiency of the circuit from point A2η (formula and numerical value). Consider the efficiency factor to be the ratio of the average power released at the load to the average power released in the entire circuit.

The power in the circuit is allocated only through resistors. Useful power is released on resistors R. Let the phase shift for this resistor be $\Delta\varphi_1$. Power across resistor a:

$$P_a = |\overline{I_a}|^2 \cos^2(\omega t - \Delta\varphi_1) R = \frac{1}{2}|\overline{I_a}|^2 R(1 + \cos(2(\omega t - \Delta\varphi_1)).$$

Since mean $\cos(2(\omega t - \Delta\varphi_1))$ over period equal 0, mean over period power:

$$\langle P_a \rangle = \frac{1}{2}|\overline{I_a}|^2 R.$$

It can be seen, that shear phase $\Delta\varphi_1$ not affects on $\langle P_a \rangle$, and means:

$$\langle P_a \rangle = \langle P_b \rangle = \langle P_c \rangle.$$

That is, the useful power:

$$P_{full} = \frac{3}{2}|\overline{I_a}|^2 R.$$

Similarly find general power in chain:

$$P_{\text{tot}} = \frac{3}{2}|\overline{I_a}|^2 (R+r).$$

By definition, give in condition:

$$\eta = \frac{P_{full}}{P_{\text{tot}}} = \frac{R}{R+r} \approx 64{,}3\%.$$

Answer:

$$\eta = \frac{R}{R+r} \approx 64{,}3\%.$$

A4 1.00 pts

Now let the load be asymmetrical: elements Z_b and Z_c are resistors with resistance R, and element Z_a is a capacitor with capacitance $C_1 = 10^{-9}$ F. Find the numerical value of the efficiency of circuit η_1 in this case. It is not necessary to obtain an analytical expression for η_1. Use the numerical data from point A2.

Capacitor impedance module:

$$|Z_a| = \frac{1}{\omega C} = \frac{1}{2\pi f C} \approx 3183\text{k}\Omega.$$

Note, that $|Z_a| \gg R$, therefore for calculation current current through capacitor one can neglect. Similarly part $A2$ write law Kirchhoff's for chain:

$$\overline{I_0} = \overline{I_b} + \overline{I_c}; \mathcal{E}_0\left(-\frac{\sqrt{3}}{2}i - \frac{1}{2}\right) = \overline{I_b}(R+r) + \overline{I_0}r; \mathcal{E}_0\left(\frac{\sqrt{3}}{2}i - \frac{1}{2}\right) = \overline{I_c}(R+r) + \overline{I_0}r \tag{3}$$

Add (2) and (3)(1). Thus find $\overline{I_0}$:

$$\overline{I_0} = -\frac{\mathcal{E}_{\prime}}{R+3r}.$$

Substitute find $\overline{I_0}$ in (2) and (3), find $\overline{I_b}$ and $\overline{I_c}$:

$$\overline{I_b} = \frac{\mathcal{E}_{\prime}}{R+r}\left(-\frac{\sqrt{3}}{2}i + \left(\frac{r}{R+3r} - \frac{1}{2}\right)\right); I$$
$$\overline{I_c} = \frac{\mathcal{E}_{\prime}}{R+r}\left(\frac{\sqrt{3}}{2}i + \left(\frac{r}{R+3r} - \frac{1}{2}\right)\right);$$

Expression for magnitude/modulus:

$$|\overline{I_0}| = \frac{\mathcal{E}_{\prime}}{R+3r} \approx 12{,}92\text{mA};$$
$$|\overline{I_b}| = |\overline{I_c}| = \frac{\mathcal{E}_{\prime}}{R+r}\sqrt{\frac{3}{4} + \left(\frac{r}{R+3r} - \frac{1}{2}\right)^2} \approx 20{,}23\text{mA};$$

Find the mean useful and the total power:

$$P_{full} = 2\frac{1}{2}|\overline{I_b}|^2 R;$$
$$P_{\text{tot}} = 2\frac{1}{2}|\overline{I_b}|^2(R+r) + \frac{1}{2}|\overline{I_0}|^2 r;$$
$$\eta_1 = \frac{2|\overline{I_b}|^2 R}{2|\overline{I_b}|^2(R+r) + |\overline{I_0}|^2 r} \approx 59{,}9\%.$$

Answer:

$$\eta_1 \approx 59{,}9\%.$$

B1 — 0.60 pts

Let a current I_0 flow in one of the square frames with side a at a given moment of time. Find the magnetic field B_0 at the center of the square created only by this frame.

First, consider a segment with current I. Let us find the field at a point located at a distance h from the line containing it. Biot-Savart-Laplace law:

$$d\vec{B} = \frac{\mu_0 I_0}{4\pi}\frac{[\vec{dl} \times \vec{r}]}{|\vec{r}|^3}.$$

Field from any small of the fragment segment field be direct equally – by Oz. Taking into account this write expression for projection field $d\vec{B}$ on Oz:

$$dB_z = \frac{\mu_0 I_0}{4\pi}\frac{dl\cos(\alpha)}{r^2}.$$

Substitute $dl = \frac{hd\alpha}{\cos^2(\alpha)}$, $r = \frac{h}{\cos(\alpha)}$:

$$dB_z = \frac{\mu_0 I_0}{4\pi h}\cos(\alpha)d\alpha;$$

$$B_z = \frac{\mu_0 I_0}{4\pi h}\int\limits_{\alpha_1}^{\alpha_2}\cos(\alpha)d\alpha;$$

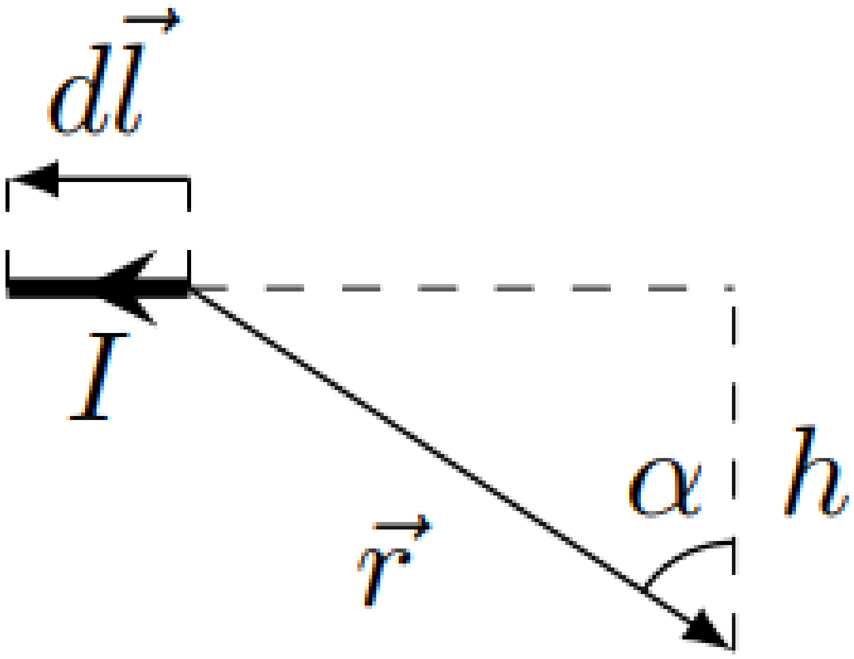
$$B_z = \frac{\mu_0 I_0}{4\pi h}(\sin(\alpha_2) - \sin(\alpha_1)).$$

For one side of the square frame $h = a/2$, $\alpha_2 = -\alpha_1 = \pi/4$. Then the margin in the center of the frame:

$$B_0 = 4\frac{\mu_0 I_0}{4\pi\frac{a}{2}}\left(2\frac{\sqrt{2}}{2}\right) = 2\sqrt{2}\frac{\mu_0 I_0}{\pi a}.$$

Answer:

$$B_0 = 2\sqrt{2}\frac{\mu_0 I_0}{\pi a}.$$



B2 1.50 pts

We will now consider the field that is created by all three frames, the currents through which are indicated in the introduction. Prove that in this case the magnetic field rotates with a constant angular velocity in the plane xy, that is, its modulus is constant and equal to B_1, and the angle with the axis $x\psi = \omega t + \psi_0$ (then $B_x = B_1\cos\psi$, $B_y = B_1\sin\psi$). Find B_1 and ψ_0. Express your answer using B_0.

We obtain the projection of the field from the three frames onto Ox:

$$B_x = B_0\cos(\omega t) + B_0\cos\left(\omega t - \frac{2\pi}{3}\right)\cos\left(\frac{2\pi}{3}\right) + B_0\cos\left(\omega t - \frac{4\pi}{3}\right)\cos\left(\frac{4\pi}{3}\right) =$$
$$= B_0\left(\cos(\omega t) - \frac{1}{2}\left(cos\left(\omega t - \frac{2\pi}{3}\right) + \cos\left(\omega t - \frac{4\pi}{3}\right)\right)\right).$$

Apply formula sum cosine:

$$B_0\left(\cos(\omega t) - \cos\left(\omega t - \pi\right)\cos\left(\frac{\pi}{3}\right)\right) = \frac{3}{2}B_0\cos(\omega t).$$

Similarly one can calculate projection total field on Oy:

$$B_y = B_0 \cos\left(\omega t - \frac{2\pi}{3}\right) \sin\left(\frac{\pi}{3}\right) - B_0 \cos\left(\omega t - \frac{4\pi}{3}\right) \sin\left(\frac{\pi}{3}\right),$$

$$B_y = \frac{\sqrt{3}}{2} 2 \sin(\omega t - \pi) \sin\left(-\frac{\pi}{3}\right) = \frac{3}{2} B_0 \sin(\omega t).$$

Both expressions lead to the answer:

$$B_1 = \frac{3}{2} B_0; \qquad \psi_0 = 0.$$

Alternative solution. Let $\vec{e_x}$, $\vec{e_y}$ be the unit vectors of the x, y axes, respectively. Let us use them to write down vector expressions for fields from three frames:

$$\vec{B_a} = B_0 \vec{e_x} \cos(\omega t);$$

$$\vec{B_b} = B_0 \left(-\frac{1}{2}\vec{e_x} + \frac{\sqrt{3}}{2}\vec{e_y}\right) \cos\left(\omega t - \frac{2\pi}{3}\right);$$

$$\vec{B_c} = B_0 \left(-\frac{1}{2}\vec{e_x} - \frac{\sqrt{3}}{2}\vec{e_y}\right) \cos\left(\omega t - \frac{4\pi}{3}\right).$$

Transition to complex amplitude:

$$\vec{B_b} = B$$

$$\vec{B_c} = B_0 \left(-\frac{1}{2}\vec{e_x} - \frac{\sqrt{3}}{2}\vec{e_y}\right) e^{i\omega t} e^{-i\frac{4\pi}{3}} = B_0 \left(-\frac{1}{2}\vec{e_x} - \frac{\sqrt{3}}{2}\vec{e_y}\right) e^{i\omega t} e^{i\frac{2\pi}{3}} . \vec{B_1} = \vec{B_a} + \vec{B_b} + \vec{B_c} = B_0 e^{i\omega t} \left(\vec{e_x} \left(1 - \frac{e^{i\frac{2\pi}{3}} + e^{-i\frac{2}{3}}}{2}\right.\right.$$

$$= B_0 e^{i\omega t} \left(\vec{e_x} \left(1 - \cos \left(\right.\right.\right.$$

Take the real part from expression above, obtain:

$$\vec{B_1} = \frac{3}{2} B_0 \left(\vec{e_x} \cos(\omega t) + \vec{e_y} \sin(\omega t)\right).$$

This implies the answer obtained earlier.

Answer:

$$B_1 = \frac{3}{2} B_0; \qquad \psi_0 = 0.$$

B3 — 0.40 pts

The small box in the center is rotated so that its normal makes an angle α with the axis x. Find the magnetic flux through it at time t. Express your answer using B_1, S, α, ω.

Answer:

$$\Phi = B_1 S \cos(\alpha - \omega t).$$

B4 — 1.00 pts

Let the inductance of the frame be L_2 and the resistance R_2. The frame rotates at a constant angular velocity ω_1, so $\alpha = \omega_1 t$ $(0 < \omega_1 < \omega)$. In the steady state, the current in the loop depends on time as

$$I(t) = A \cos(\omega_2 t - \varphi_2).$$

Find A, ω_2 and φ_2. Express answer through parameter of the frame, B_1, ω, ω_1.

The total magnetic field flux through the frame is created by the rotating field and current in the frame I:

$$\Phi_t = \Phi + L_2 I.$$

Total induced EMF in the frame with current I:

$$\mathcal{E} = \dot{\Phi}_t = -\dot{\Phi} - L_2\dot{I}.$$

From the law Ohm's for of the frame $IR_2 = \mathcal{E}$ obtain

$$-\dot{\Phi} = L_2\dot{I} + IR_2$$

This equation is formally equivalent to equation for RL-chain with source variable voltage $-\dot{\Phi}$. Be write oscillation in complex form:

$$\Phi = B_1 S e^{i(\omega-\omega_1)t}.$$

Since flux/flow change with frequency $\omega_2 = \omega - \omega_1$, with that same frequency be change and current. induced EMF in the frame:

$$\mathcal{E}(t) = -\dot{\Phi} = -i\omega_2 B_1 S e^{i\omega_2 t}.$$

Substitute dependence current from time in form $I = \overline{I}e^{i\omega_2 t}$, obtain

$$(i\omega_2 L_2 + R_2)\overline{I} = -i\omega_2 B_1 S; \overline{I} = -\frac{i(\omega-\omega_1)B_1 S}{i(\omega-\omega_1)L_2 + R_2}.$$

Hence we get answer

$$\omega_2 = \omega - \omega_1$$

$$A = |\overline{I}| = \frac{B_1 S(\omega-\omega_1)}{\sqrt{R_2^2 + (\omega-\omega_1)^2 L_2^2}};$$

$$\varphi_2 = \pi - \arctan\left(\frac{R_2}{(\omega-\omega_1)L_2}\right).$$

Answer:

$$\omega_2 = \omega - \omega_1;$$

$$A = \frac{B_1 S(\omega-\omega_1)}{\sqrt{R_2^2 + (\omega-\omega_1)^2 L_2^2}};$$

$$\varphi_2 = \pi - \arctan\left(\frac{R_2}{(\omega-\omega_1)L_2}\right).$$

B5 — 1.50 pts

In order to ensure more uniform rotation of the rotor, instead of one frame we will use three frames, rotated relative to each other by $120°$ around the z axis (that is, if the position of one frame is specified by the angle α, the positions of the remaining frames are specified by the angles $\alpha_2 = \alpha + 2\pi/3$, $\alpha_3 = \alpha + 4\pi/3$). The frames are isolated from each other; their mutual induction can be neglected. Prove that the projection onto the z axis of the moment of forces acting on the structure of three frames is constant. Find its value M_z. Express your answer in terms of the frame parameters, B_1, ω, ω_1.

The magnetic flux of the rotating magnetic field through the ith frame ($i = 1, 2, 3$) is equal to $\Phi_i = B_1 S\cos(\omega_2 t - \alpha_{0i})$ ($\alpha_{01} = 0$, $\alpha_{02} = 2\pi/3 = \Delta\alpha$, $\alpha_{03} = 4\pi/3 = 2\Delta\alpha$). Thus, the magnetic flux through the second and third frames is delayed by $2\pi/3$ and $4\pi/3$. Then the currents through these frames are delayed by the same phase. Magnetic moment of a frame with current I and area S is $m = IS$. Then on the i-th frame the torque acts

$$\vec{M}_i = [\vec{m}_i \times \vec{B}].$$

in projection on Oz:

$$M_{iz} = m_i B_1 \sin(\psi - \alpha_i).$$

Thus (notation for current take from previous part):

$$M_z = B_1 AS(\sin(\omega_2 t)\cos(\omega_2 t - \varphi_2) + \sin(\omega_2 t - \Delta\alpha)\cos(\omega_2 t - \varphi_2 - \Delta\alpha) + \\ + \sin(\omega_2 t - 2\Delta\alpha)\cos(\omega_2 t - \varphi_2 - 2\Delta\alpha)).$$

Therefore

$$M_z = \frac{1}{2}B_1 AS\big(\sin\varphi_2 + \sin(2\omega_2 t - \varphi_2) + \sin\varphi_2 + \sin(2\omega_2 t - \varphi_2 - 2\Delta\alpha) \\ + \sin\varphi_2 + \sin(2\omega_2 t - \varphi_2 - 4\Delta\alpha)\big).$$

Sum three sine, depend from time, becomes in 0, therefore

$$M_z = \frac{3}{2}B_1 AS\sin\varphi_2.$$

Substitute φ_2, A, obtain

$$\sin\varphi_2 = \sin\left(\arctan\frac{R_2}{(\omega-\omega_1)L_2}\right) = \frac{R_2}{\sqrt{R_2^2 + (\omega-\omega_1)^2 R_2^2}}.$$

Alternative solution. Let's find the total moment of three frames with current. This problem is mathematically equivalent to the problem of finding the total magnetic field from three external current-carrying loops. Indeed, the magnetic moment of each subsequent frame is rotated by 120° degrees relative to the magnetic moment of the previous frame, and is delayed in phase by $2\pi/3$. Let us first consider the problem in the reference frame of a rotating frame. Then if $m_0 = AS$ is the amplitude of the magnetic moment from one frame, then the value of the total magnetic moment is $m_1 = \frac{3}{2}m_0$, and the magnetic moment rotates with an angular velocity ω_2. In this case, the initial phase of the magnetic moment is equal to $-\varphi_2$, since all currents lag relative to the magnetic field by this phase. Since the frames themselves rotate with an angular velocity of ω_1, relative to the laboratory frame of reference, the magnetic moment rotates with an angular velocity of $\omega_1 + \omega_2 = \omega$. The angular velocities of the magnetic moment and the magnetic field coincide, so the angle between them remains constant. It follows that the moment of forces is also constant and equal to

$$M_z = \frac{3}{2}m_0 B_1 \sin\varphi_2 = \frac{3}{2}ASB_1\sin\varphi_2.$$

Thus, we obtain the same answer.

Answer:

$$M_z = \frac{3}{2}B_1^2 S^2 \frac{(\omega-\omega_1)R_2}{(\omega-\omega_1)^2 L_2^2 + R_2^2}.$$

B6 1.00 pts

Construct a qualitative graph of the dependence $M_z(\omega_1)$ in the frequency range $0 < \omega_1 < \omega$. Specify the coordinates of the characteristic points. Consider that $R_2/L_2 < \omega$.

Let's rewrite the expression for the moment of forces in the form

$$M = \frac{3}{2}B_1^2 S^2 \frac{R_2}{L_2^2}\frac{1}{(\omega-\omega_1) + \frac{R_2^2}{L_2^2}\frac{1}{(\omega-\omega_1)}}.$$

The maximum moment corresponds to the minimum of the denominator

$$(\omega-\omega_1) + \frac{R_2^2}{L_2^2}\frac{1}{(\omega-\omega_1)}.$$

Take derivative by $\omega - \omega_1$, we obtain that the extremum corresponds to

$$\omega - \omega_{1max} = \frac{R_2}{L_2};$$

$$\omega_{1max} = \omega - \frac{R_2}{L_2}.$$

Maximum value moment $M_{max} = \frac{3}{4}\frac{B_1^2 S^2}{L_2}$. Curiously note, that M_{max} not depend from R_2. For $\omega_1 = 0$

$$M_0 = \frac{3}{2} B_1^2 S^2 \frac{\omega R_2}{\omega^2 L_2^2 + R_2^2}.$$